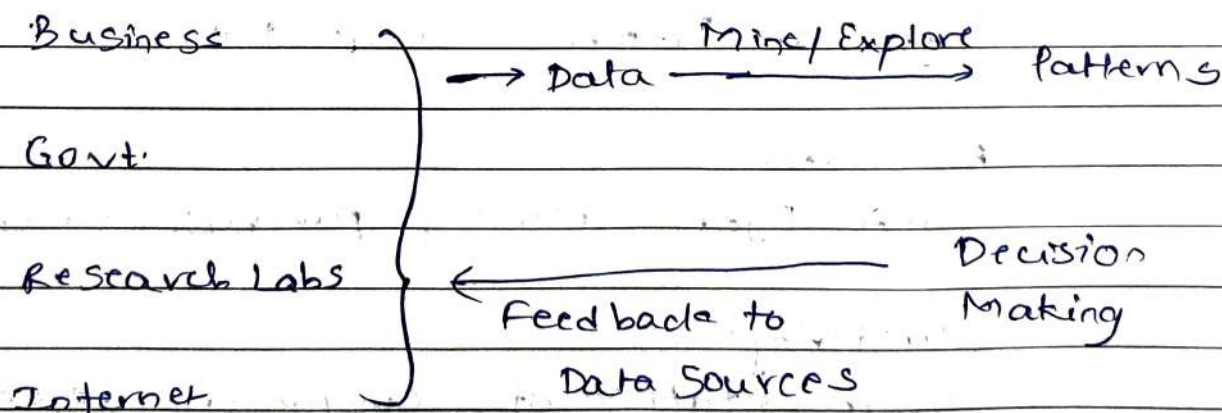
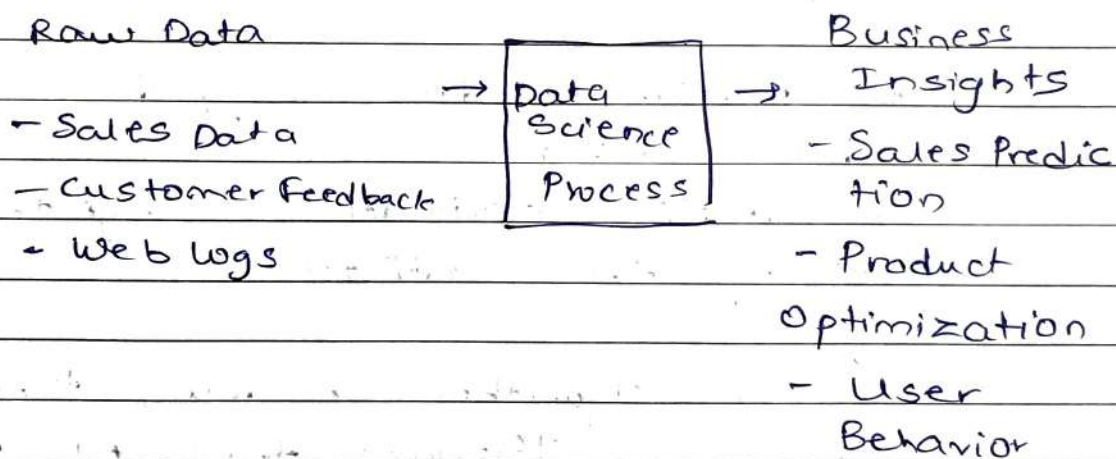


19-2-25

Sujit Class 1.

Midsem syll till today. Next class for Q2.
Data Science & Data mining.

Data science is an interdisciplinary academic field that use statistics, Scientific computing, Scientific methods, processing, scientific visualization, algorithms & systems to extract or extrapolate knowledge from potentially noisy, structured or unstructured data. (wiki)



Data Mining

Data mining is the process of discovering valuable insights, patterns & information from vast data sets by employing various techniques, algorithms, & tools.

Data mining is a technology that blends traditional data analysis methods with sophisticated algorithms for processing large volumes of data.

In particular, data mining draws upon ideas, such as

- sampling, estimation, & hypothesis testing from statistics

- Search algorithms, modeling techniques, & learning theories from AI, pattern recognition & ML

Types of Patterns : (Data Mining)

Associations:

Coffee Buyers usually purchase sugars.

Clustering:

Segmentation of customers requiring different promotion strategies.

Classification:

Bank customer is applying for loan, will applicant be defaulter?

1	Bread	Butter	Milk
2	"	"	Salt
3	"	"	Milk
4	"	"	Salt

$$\begin{array}{l} \text{Bread} \xrightarrow{x} \text{Butter} \xrightarrow{y} 100\% \\ (\text{Bread}, \text{Butter}) \rightarrow \text{Milk} \quad 50\% \\ (\text{Bread}, \text{Butter}) \rightarrow \text{Salt} \quad 50\% \end{array}$$

Association Rules

Antecedent, consequent

AR: $X \rightarrow Y$ $\{x_1, x_2, \dots, x_k\} \rightarrow \{y_1, y_2, \dots, y_k\}$

$|X|$ need not be same as $|Y|$

itemset is list of all items in the antecedent & consequent i.e. $X \cup Y$

Support of Ass. Rule

1. $\text{Supp}(AR) = \text{Prob}(X \cup Y)$

Does not differentiate b/w $X \rightarrow Y$ & $Y \rightarrow X$

//_

or # transactions containing $X \cap Y$

transactions in D (total)
↓
Database

This is one of the 3 important measures called 'association rule'

$$\text{Support (AR)} = \text{Prob}(X \cap Y)$$

Say on 100 (is total)

Milk	Both	Tooth Brush
70	10	4

$X \rightarrow Y$
TB $\rightarrow$ Milk

Itemset : $\{X, Y\}$

= $\{TB, Milk\}$

X : Toothbrush

Y : Milk

$$\text{Supp (AR)} = \frac{P(X \cap Y)}{100} = \frac{10}{100} = 0.1$$

2. $\text{Confidence (AR)} = \text{Prob}(Y | X)$

transactions containing $X \cap Y$

transactions in X

$$\text{Confidence (AR)} = \frac{10}{14} \approx 0.71$$

(We work with empirical mean
(versions of Supp/conf))

This association is good because $P(Y)$ is very high $\cdot 0.8$.

$$3. \text{ Lift (AR)} = \frac{\text{Prob}(Y|X)}{\text{Prob}(Y)}$$

If $\text{Lift} > 1$ Good Association
Else can be ignored.

Empirical:

$$\frac{\# \text{trans cont. } X \cap Y}{\left(\frac{\# \text{trans in } X}{\# \text{trans in } Y} \right) \cdot \# \text{trans in } D}$$

$$\text{Lift (AR)} = \frac{0.7}{0.8} \approx 0.87$$

If all 3 are good, association good.

If there are total of 10 items:

Total # of AR excluding $x \rightarrow x$ & $x \rightarrow y$ & $y \rightarrow y$

$$\binom{10}{1} \times \left[\binom{9}{1} + \binom{9}{2} + \dots + \binom{9}{9} \right]$$

$$2^9 - 1$$

$$+ \binom{10}{2} \times \left[\binom{8}{1} + \binom{8}{2} + \dots + \binom{8}{8} \right]$$

$$+ \binom{10}{3} \times \dots$$

$$= 57002$$

$$n = 10$$

$$\sum_{i=1}^n \binom{n}{i} \times [2^{n-i} - 1]$$

Grows faster than 2^n

Applications of Ass. Rules:

Market Basket Analysis:

- People who have bought HP also bought PJ.

Recommendation Systems

Fraud Detection

Healthcare & Medical Research

— Allergy to latex rubber usually co-occurs with allergies to banana & tomato.

Census analysis:

Immigrants are usually male.

Sports:

Chess.

Exercise:

Transaction id	Items
1	Tomato (T), Potato (P), Onion (O)
2	T, P, Brinjal (B), Pumpkin (P)
3	T, P, O, Chilly (C)
4	Tamarind, Lemon

AR: Tom, Pot $\rightarrow$ Onion

X: T, P

Y: O

AR: X $\rightarrow$ Y

Itemset: {T, P, O}

$$\text{Supp (AR)} = \frac{2}{4} = 0.5$$

$$\text{Confidence} = \frac{2}{3} = 0.66$$

$$\text{Lift (AR)} = \frac{2/3}{4/2} = \frac{4}{3}$$

P(Onion) $\uparrow$ if we have Tom & Pot
So good association rules.

Challenges in Identifying ARs.

How many association rules are possible with 10 diff items?

57002

This increases exponentially.

//_

If we have $Y \rightarrow X$ or $X \rightarrow Y$
Supp is same

What about $\text{Lift}(X \rightarrow Y)$ and $\text{Lift}(Y \rightarrow X)$
Yes.

$$\text{Lift}(X \rightarrow Y) = \frac{\#t(X \cap Y)}{\#t(X)}$$
$$\frac{\#t(Y)}{\#t(D)}$$

$$= \frac{\#t(X \cap Y)}{\#t(X)} \times \frac{\#t(D)}{\#t(Y)}$$

$$= \frac{\#t(X \cap Y)}{\#t(Y)} \times \frac{\#t(D)}{\#t(X)}$$

$$= \text{Lift}(Y \rightarrow X)$$

Still 2^{10} itemsets need to be considered
if they meet reasonable support.

Apriori's Algorithm:

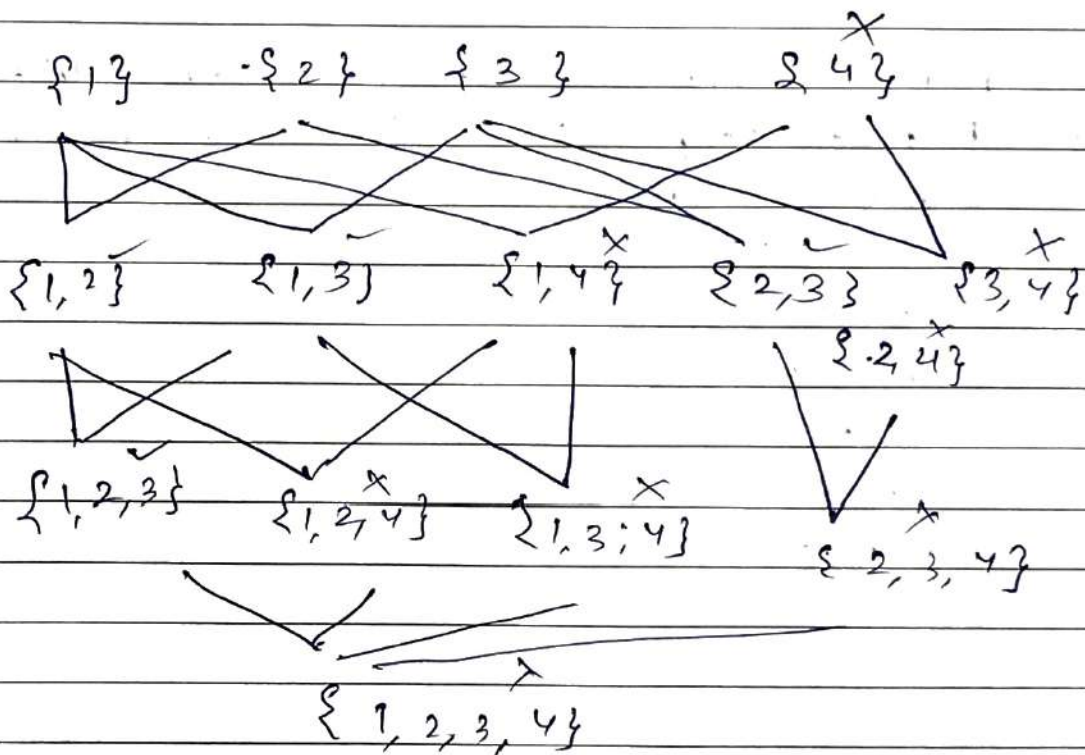
Invented by Rakesh Agrawal &
Ramakant Srikant (1994)

Can we speed up pure brute force?

Apriori acknowledges prior knowledge

If any itemset is not frequent, its
Superset cannot be frequent

An itemset can be frequent only if all
its subsets are frequent.



Start with singleton & consider subsets
not already discarded.

How does it work?

- 0: create 1 size frequent itemsets list
that meet threshold support, $k=1$
- 1: Expand itemsets list

From k sized itemsets list combine overlapping sets to $k+1$ size itemsets list

- 2: Prune the expanded itemsets list using apriori property ; $k = k+1$
- 3: Remove infrequent itemsets from list
- 4: Repeat step 1, 2, 3 till no further expansion possible.

1, 2, 3, 4

1, 2, 4

1, 2

2, 3, 4

2, 3

3, 4

2, 4

Suppose the above is AB.

So based on freq. all atoms remain

Now. we discard $\{1, 3\}$

Iter. 1
 $\{1\}$ iter 2 $\{1, 2\}$ $\{1, 3\}$ $\{1, 4\}$

$\{2\}$ $\{2, 3\}$

$\{3\}$ $\{2, 4\}$

$\{4\}$ $\{3, 4\}$ occur 3 times.

~~iter 3:~~

Any set containing 1, 3, 1, 4 discard

$\{1, 2, 3\}$ x

$\{1, 2, 4\}$ x

$\{2, 3, 4\}$